

Name: sivaramakrishnan R

Branch: M.E cyber security

CRYPTOGRAPHY ASSIGNMENT – 1

* From my analysis these are application that's involving the CIA concept

- 1) **Why should I choose these applications** in my perspective I really want to show our development of country and every person want to know how our government of India making many digital platforms so make all the basic things in easier ways and building with more secured terms so, I mentioned all the Indian related applications.

SNO	Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
1	(Aadhaar) UIDAI	Confidentiality	We already know about our Aadhar web. It's mainly used for encryption.	HTTPS / TLS	communication between the user and UIDAI services and helping prevent unauthorized persons from reading sensitive
2	IRCTC	Availability	Mainly used to booking so, it controls load balancing and backup systems	Web servers, load balancing, backup infrastructure	ticket booking and other online services available even during high traffic or system failures.
3	NPCI / UPI	Integrity	Used for transactions comes under the integrity mechanism.	UPI security mechanisms, cryptography, HTTPS/TLS	ensure that payment and transaction information is protected from unauthorized modification while being processed and transmitted.

4	India Post online services	Confidentiality	it is used for User identity verification so, encryption mechanism performed	HTTPS / TLS	information exchanged between users and India Post online services by encrypting communication reducing the risk of sensitive information being intercepted.
5	ECI Voter Services (Election commission of india)	Integrity	This is mainly used for Data validation and access control	HTTPS / TLS, cryptographic integrity mechanisms	To protect every single voter-related information from unauthorized modification and ensures that information exchanged with the service is transmitted securely.

2) real-world applications, investigate the security controls used to protect users and data.

- Let me explain about what are the applications are used for high security protection.

1) **Industrial Management Systems** : The main purpose of an industrial management is to make sure that the machines and processes are working correctly and safely.

Example: an attacker may change a temperature setting in a factory or switch off an important machine.

- confidentiality they uses access control, user authentication, VPNs, and encryption.
- availability industries need their to keep running continuously.so, They can use backup servers.
- integrity the system needs to make sure that the commands and data are not changed by an unauthorized person

2) **Airport Management System:** An Airport Management System is used to manage different activities inside an airports.

- confidentiality airports use a secure communication because many threats come from the hackers.
- In integrity airport must make sure that flight schedules, passenger details and other important records are not changed without authorization.
- Availability The airport management needs to work continuously because even a small system failure can cause delays and affect many passengers.

3) **Digi -Locker:** it is a digital platform that allows people to store their personal important documents.

- Confidentiality Users need to authenticate themselves before accessing their documents.
- integrity the system needs to make sure that documents are genuine and have not been modified.